Rishi Raj

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Education

Newton School of Technology, Rishihood University — Aug 2024 – Jun 2028
B.Tech in Computer Science & Artificial Intelligence — *Third-Year Student*

Sainik School Tilaiya — Aug 2017 – Mar 2024
Class 6 – Class 12

Experience

Maintainer & LFX Mentee – Meshery (CNCF) — Feb 2026 – Oct 2026
Maintainer across Meshery repos; LFX Mentee: Models Support for OCI Registries (2026 Term 2) — *Remote*

- Authored **100+ merged PRs**; reviewed **300+** and merged **100+** as a maintainer across meshery, meshery/schemas and sistent, keeping shared interfaces consistent across repos
- Built OCI-registry support for Meshery's model registry, implementing **ORAS**-based push/pull workflows so models ship as standard OCI artifacts (manifests, blobs, digests, media types)
- Led the permission-shield migration: centralized RBAC permission checks into the shared Sistent library, so Meshery UI, Meshery Cloud and Kanvas pass only a permission key and action
- Fixed a recurring Registry-page crash spanning multiple dependent repos; onboarded new contributors, triaged issues and improved contributor docs

Projects

Quiver – Vector Database from Scratch | *Go, HNSW, ARM64 NEON, WAL, HTTP API* — Live | GitHub

- Built a vector database in Go with zero third-party dependencies (8.7k lines): HNSW index, write-ahead log, snapshots, filtered search, HTTP API and a web UI
- Reached **0.99 recall@10 at 3,849 queries/sec per core** on the SIFT1M benchmark (1M vectors), about 75% of Meta's FAISS and 20% faster than hnswlib on the same hardware
- Wrote ARM64 NEON assembly distance kernels, **2.5x faster** than the compiled Go version
- Implemented group commit in the write-ahead log, raising durable write throughput **35x** (420 to 14,900 writes/sec); verified zero acknowledged writes lost across repeated kill -9 crashes
- Deployed a live semantic search demo over 20,000 news articles, with in-browser query embedding and sub-2 ms results

HELIX – Autonomous AI News Pipeline | *Python, LLMs (Groq), PostgreSQL, Next.js, Prisma* — Live | GitHub

- Ingests ∼400 articles/day from 13 sources, summarizes them with LLMs and ranks them per subscriber, running unattended on a GitHub Actions cron
- Sends each subscriber a personalized daily email, auto-posts top stories as an Instagram carousel, and serves a Next.js dashboard with Clerk auth and Prisma
- Cut query time **8.5s → 3.7s** by replacing a ∼32k-row in-memory scan with a SQL anti-join and indexes; added 106 offline tests, CI and alerts for zero-delivery runs
- Traced a silent production outage to 3 root causes, restoring ingest from 5 to ∼420 articles/run; enforced Free/Pro/Admin access server-side, closing a flaw that exposed subscriber emails

Technical Skills

Languages: Go, Python, TypeScript, JavaScript, SQL
Backend & Systems: HNSW, WAL / Crash Recovery, SIMD (ARM64 NEON), Node.js, FastAPI, Prisma
Databases: PostgreSQL, MongoDB, MySQL
Cloud & DevOps: Docker, Kubernetes, GitHub Actions, AWS, Vercel, OCI Registries / ORAS, Prometheus, Grafana
Frontend: React.js, Next.js, Tailwind CSS

Achievements

- Finalist, Meta × Hugging Face OpenEnv Hackathon
- CNCF LFX Mentee (2026 Term 2); Certified Meshery Contributor